score_sjsdm_joint_tuning_predictions.R

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`score_sjsdm_joint_tuning_predictions`
Score Joint sjSDM Tuning Predictions

Description

Calculates the held-out joint sjSDM negative log likelihood and probability diagnostics used by regularization tuning.

Usage

```
score_sjsdm_joint_tuning_predictions(
  object = NULL,
  data_test_input = NULL,
  data_observed = NULL,
  data_predicted = NULL,
  epsilon = 1e-06,
  n_likelihood_draws = 20L,
  score_seed = 900723L
)
```

Arguments

`object` Fitted ‘sjSDM‘ object containing the model likelihood callback, formulas, and fit settings.

`data_test_input` Fold-local test input containing ‘data_abiotic_to_fit‘ and, for spatial models, ‘data_spatial_to_fit‘.

`data_observed, data_predicted` Aligned binary observations and marginal predicted probabilities.

`epsilon` Probability clipping tolerance passed to [score_sjsdm_tuning_predictions()].

`n_likelihood_draws`
: Positive integer number of stochastic likelihood evaluations to average. Defaults to '20L', matching 'sjSDM::sjSDM_cv()'.

`score_seed`
: Non-negative integer used for deterministic stochastic likelihood draws.

Details

Test design matrices are reconstructed from the fitted model formulas. Likelihood sampling and batch size are read from the fitted model settings. The caller's R and available PyTorch random-number states are restored.

Value

One-row tibble with retained taxa, response-value count, total joint negative log likelihood, joint loss per response, and macro AUC.

Examples

```
## Not run:
score_sjsdm_joint_tuning_predictions(
  object = mod_fit,
  data_test_input = list_test_input,
  data_observed = data_test_observed,
  data_predicted = data_test_predicted
)

## End(Not run)
```

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